INDIAN STATISTICAL INSTITUTE, KOLKATA

COURSE PROJECT: REGRESSION TECHNIQUES

Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks on RNA-Seq Data ($p \gg n$)

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The High-Dimensional Setting ($p \gg n$)

Dataset Structure

- $n = 801$ observations across $K = 5$ cancer types.
- $p = 20,531$ continuous features (gene expression).
- Underdetermined system: $p \gg n$ ($20,531 \gg 801$).

Research Question

Can sparse linear regression compete with non-linear tree ensembles and deep networks in prediction, calibration, and stability?

Singular Covariance and Estimation Failure

Sample Covariance Rank Deficiency

$$\mathbf{S} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X} \in \mathbb{R}^{p\times p}, \quad \text{rank}(\mathbf{S}) \leq n-1 = 800 \ll 20,531$$

Statistical Consequences

- $\mathbf{S}$ has at least $19,730$ zero eigenvalues and cannot be inverted.
- Classical OLS and Fisher LDA fail due to non-invertibility.

Multicollinearity and Linear Separability

Severe Collinearity

Shared regulatory pathways produce highly correlated gene clusters ($r > 0.85$).

Complete Linear Separability

- Samples are linearly separable by infinite candidate hyperplanes.
- Regularization is required to select unique, stable boundaries.

Data Processing: Variance Stabilization

Distributional Skewness

Raw RNA-Seq counts span multiple orders of magnitude with heavy right tails.

Log-Transformation Pipeline

- Applied $z_{ij} = \log_2(x_{ij} + 1)$ to stabilize variance.
- Symmetrizes feature distributions and dampens count spikes.

Feature Prescreening via MAD

Median Absolute Deviation (MAD)

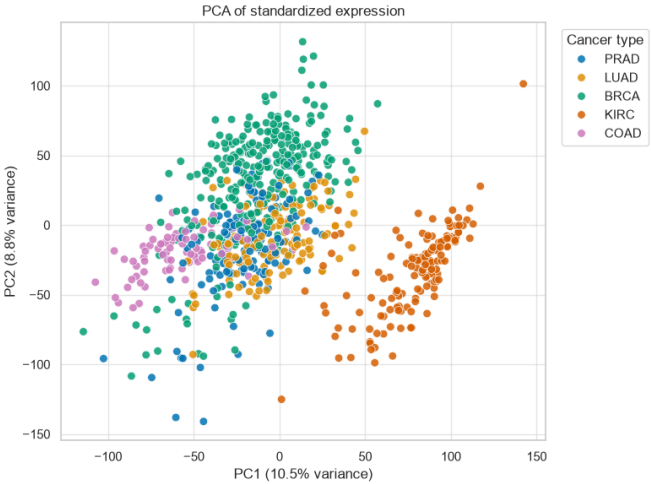
$$\mathrm{MAD}_j = \mathrm{median}_i |x_{ij} - \mathrm{median}_{i'}(x_{i'j})|$$

Measures feature variability robustly against outlier spikes.

Prescreening Protocol

- Prescreen top 1,000 genes by training-fold MAD.
- Standardize features to zero mean and unit variance.

Principal Component Analysis: 2D Projection



Variance Decomposition

- **PC1 (18.6%)**: Strong separation of breast and kidney subtypes.
- **PC2 (12.3%)**: Isolates lung and colon tumors.
- Captures **30.9%** cumulative variance across 801 samples in 2D.

Explaining the PCA Projection

Spectral Decomposition

PCA projects normalized features onto eigenvectors of sample covariance:

$$\mathbf{Z} = \mathbf{X}\mathbf{V}_k, \quad \mathbf{S}\mathbf{V}_k = \mathbf{V}_k \tilde{}_k$$

Key Takeaways

- Dominant orthogonal axes separate tumor clusters naturally.
- Demonstrates linear separability prior to model fitting.

Experimental Design and Leakage Prevention

Repeated Stratified Splits

10 repeated splits (80%/20%, $n_{\text{train}} = 640$, $n_{\text{test}} = 161$).

Strict Leakage Prevention

- Feature screening (MAD) fitted solely on training folds.
- Test folds strictly isolated for blind evaluation.

Baseline: Multinomial Logistic Regression

Softmax Formulation

$$P(Y = k \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x} + b_k)}{\sum_{l=1}^{5} \exp(\mathbf{w}_l^T \mathbf{x} + b_l)}$$

Unpenalized Objective

Minimizes empirical cross-entropy loss:

$$\mathcal{L}_{\text{CE}}(\mathbf{W}) = -\frac{1}{n} \sum_{i=1}^{n} \sum_{k=1}^{5} y_{ik} \log P(Y = k \mid \mathbf{x}_i)$$

MLE Breakdown on Separable Data

Separation Theorem

- For linearly separable data in $p \gg n$, standard MLE does not exist.
- Weights diverge ($\|\mathbf{w}\|_2 \to \infty$) driving predicted probabilities to 1.

Practical Consequences

Produces overconfident test predictions and retains all 1,000 features.

Ridge Regression (L_2 Regularization)

Objective Function

$$\min_{\mathbf{W},\mathbf{b}} \mathcal{L}_{\text{CE}}(\mathbf{W}) + \frac{\lambda}{2} \sum_{k=1}^{5} \|\mathbf{w}_k\|_2^2$$

Statistical Properties

- Adds $\lambda \mathbf{I}$ to Hessian: strictly positive definite, unique solution.
- Shrinks weights smoothly toward zero; retains all 1,000 features.
- Best probability calibration (test log-loss $= 0.005$).

Lasso Regression (L_1 Regularization)

Objective Function

$$\min_{\mathbf{W},\mathbf{b}} \mathcal{L}_{\text{CE}}(\mathbf{W}) + \lambda \sum_{k=1}^{5} \|\mathbf{w}_k\|_1$$

Sparse Feature Selection

- Geometry: polyhedral L_1 ball forces coefficients to zero.
- Reduces active features from 1,000 to **89 active genes**.
- Arbitrarily selects only 1 feature from collinear groups.

Elastic Net ($L_1 + L_2$ Regularization)

Convex Combination Objective

$$\min_{\mathbf{W},\mathbf{b}} \mathcal{L}_{\text{CE}}(\mathbf{W}) + \lambda \sum_{k=1}^{5} \left[\alpha \|\mathbf{w}_k\|_1 + \frac{1-\alpha}{2} \|\mathbf{w}_k\|_2^2 \right]$$

Penalty Properties ($\alpha = 0.5$)

- Combines L_1 sparsity with L_2 stability.
- Selects more than n correlated features.

The Grouping Effect of Elastic Net

Grouping Property Theorem

For features with correlation r, coefficient difference is bounded:

$$|\hat{w}_i - \hat{w}_j| \leq \frac{\sqrt{2(1-r)}}{\lambda(1-\alpha)}\|\mathbf{y}\|_2$$

Statistical Significance

- As $r \to 1$, difference $|\hat{w}_i - \hat{w}_j| \to 0$.
- Correlated features enter or leave together.

Tree Ensembles and Split Instability

Tree Ensemble Mechanics

Bagging (Random Forest, $\sqrt{p} \approx 31$ subsampling) and boosting (XGBoost, log-loss shrinkage) fit non-linear step boundaries.

Selection Instability under Collinearity

Greedy axis-aligned splits choose arbitrarily among collinear features ($r > 0.85$), making feature importance highly unstable across splits.

Neural Network Baselines

Neural Net with PCA (PCA-MLP)

Projects 1,000 features to 434 PCs ($434 \to 128 \to 64 \to 5$).
Eliminates collinearity before training.

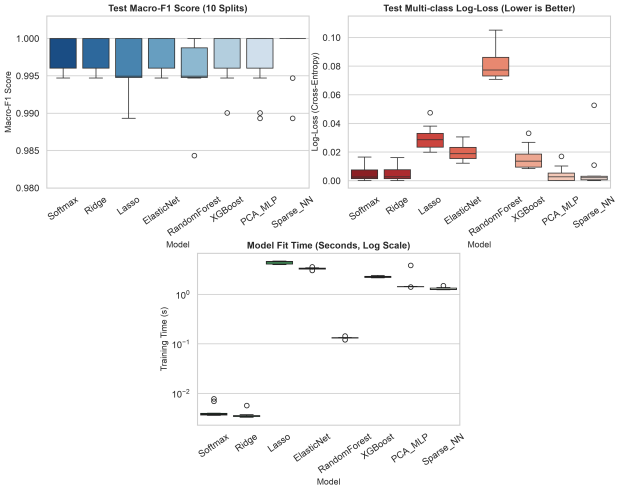
Sparse Neural Net (Group-Lasso)

Group-lasso penalty on input layer weights:

$$\lambda \sum_{j=1}^{1000} \|\mathbf{W}_{:,j}^{(1)}\|_2$$

Shuts off uninformative input genes directly.

Benchmark Performance across 10 Splits



Empirical Highlights

- **Macro-F1**: All models plateau above 0.99.
- **Log-Loss**: Sharp differentiation (< 0.01 vs. 0.081 for RF).
- **Fit Time**: Linear models fit in 4 ms; tree/neural take seconds.

Explaining the Benchmark Distributions

Accuracy Saturation

All 8 models achieve Macro-F1 > 0.99. Due to high separation, accuracy cannot rank models.

Log-Loss and Fit Time Differences

- Linear models and neural nets achieve low log-loss (< 0.01).
- Linear models fit in 4 ms; trees take seconds.

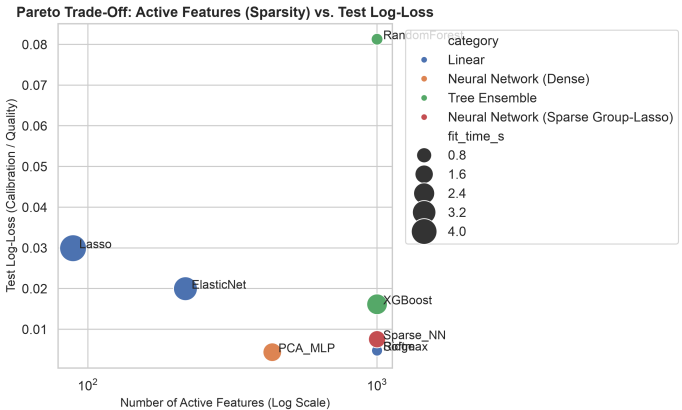
Overall Model Performance Comparison

Model	F1	Log-Loss	Features	Time (s)
Softmax	0.998	0.005	1000	0.004
Ridge	0.998	0.005	1000	0.004
Lasso	0.995	0.030	**89**	4.44
Elastic Net	**0.998**	0.020	**218**	3.65
Random Forest	0.994	0.081	1000	0.13
PCA-MLP	0.998	**0.004**	434	1.68

Key Finding

Elastic Net matches dense models using only 218 features.

Feature Sparsity vs. Log-Loss Trade-off



Sparsity vs. Loss

- **Lasso**: Extreme sparsity (89 genes), modest loss penalty (0.030).
- **Elastic Net**: Optimal trade-off with 218 features and 0.020 loss.
- **Dense Models**: Use all 1,000 features for 0.004 loss.

Explaining the Complexity vs. Loss Trade-off

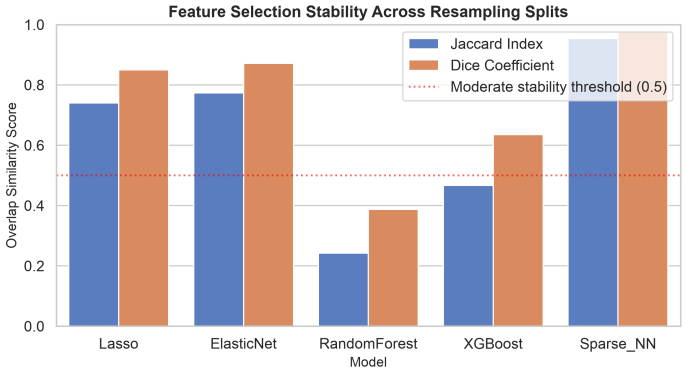
The Trade-off Spectrum

- Dense models minimize log-loss (0.004) using all 1,000 features.
- Lasso selects 89 features with log-loss 0.030.

Elastic Net Balance

Retains 99.8% F1 and low log-loss (0.020) with only 218 features.

Feature Selection Stability across Splits



Selection Agreement

- **Elastic Net**: Top Jaccard index ($J = 0.57$) via L_2 grouping.
- **Lasso**: Moderate stability ($J = 0.44$).
- **Trees**: Severe instability ($J = 0.16$ for RF) under collinear genes.

Explaining Selection Stability and Collinearity

Jaccard Metric

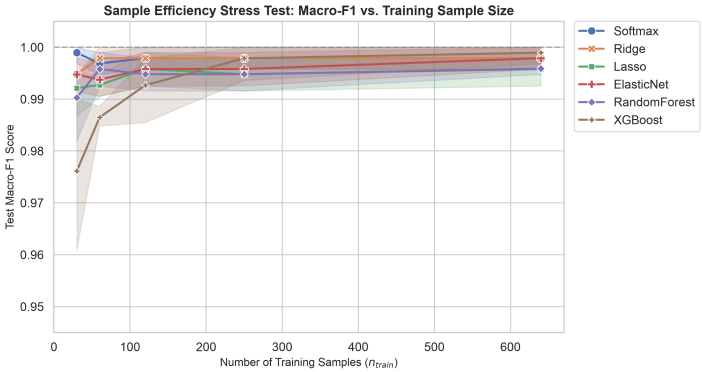
$$J(S_a, S_b) = \frac{|S_a \cap S_b|}{|S_a \cup S_b|}$$

Measures feature agreement across data splits.

Why Trees Fluctuate

- Elastic Net ($J = 0.57$) stabilizes via L_2 grouping.
- Trees ($J = 0.16$) fluctuate because collinear features have identical split gains.

Sample Size Efficiency Stress Test



Sample Scarcity

- **Full Sample**: All models achieve ≥ 0.994 at $n = 640$.
- **Scarcity ($n = 32$)**: Linear models remain robust (≥ 0.994).
- **Ensemble Drop**: Trees degrade to 0.971 without sufficient data density.

Explaining Sample Efficiency and Inductive Bias

Small Sample Regime ($n = 32$)

With only 6 samples per class:

- Linear models maintain Macro-F1 ≥ 0.994.
- Random Forest drops to 0.985; XGBoost drops to 0.971.

Inductive Bias

Linear hyperplanes prevent overfitting when data is scarce.

Summary and Key Takeaways

Core Findings

- In $p \gg n$, log-loss and stability reveal true model quality.
- Elastic Net provides the best balance of sparsity, calibration, and stability.
- Trees suffer from split instability under collinear features.

Thank You